

Gan Minghao

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EDUCATION

Sun Yat-sen University <i>B.Eng. in School of Software Engineering</i> – Academic Records: GPA 4.2/5.0, Rank: 10/231 (Top 5%)	Sep. 2023 – Jun. 2027 (Expected) Guangzhou/Zhuhai, China
The University of Hong Kong <i>Exchange Student in School of Computing and Data Science</i>	Jan. 2026 – May 2026 Hong Kong

RESEARCH EXPERIENCE

NAISS Lab, The University of Hong Kong <i>Research Assistant, advised by Prof. Shinan Liu</i> – Mixed-precision KV cache serving for long-context inference. Existing mixed-precision KV cache systems rely on custom attention kernels and lag behind vLLM/SGLang. Building a serving backend with a slab-based memory pool and an attention wrapper that reuses native kernels per precision and merges outputs via log-sum-exp, deployed as a vLLM/SGLang plugin supporting multiple KV cache quantization algorithms. – Other explorations. Cross-architecture and positional KV cache reuse to cut redundant prefill in multi-turn agent interactions; scheduling that reclaims KV cache idle periods caused by tool-call latency.	Jan. 2026 – Present Hong Kong
InPlus Lab, Sun Yat-sen University <i>Research Intern, advised by Prof. Wuhui Chen</i> – KAIROX: GPU–CPU hybrid LLM inference (USENIX OSDI '26). Sparsity-based edge inference systems split neurons between GPU and CPU statically, leaving hardware underutilized. Co-designed bottleneck-aware neuron caching and an asynchronous hybrid pipeline with runtime neuron transfer, and implemented the sparse CUDA kernels, reaching up to 7.57× speedup over llama.cpp and 6.35× over sparse baselines.	Nov. 2024 – Dec. 2025 Guangzhou, China

PUBLICATIONS

KAIROX: Adaptive GPU–CPU Hybrid LLM Inference via Online Neuron Balancing	USENIX OSDI '26
Yapeng Jiang, Minghao Gan , Zicong Hong, Wuhui Chen, Junyuan Liang, Yue Yu, Meng Guo, Zibin Zheng	

SCHOLARSHIP & AWARDS

National Scholarship for Undergraduates (twice), Ministry of Education of China	2026, 2023
First Prize, Outstanding Student Scholarship (twice), Sun Yat-sen University	2026, 2023
Fung Scholarship , Fung Foundation & The University of Hong Kong	2025
Second Prize, Outstanding Student Scholarship , Sun Yat-sen University	2024
Special Scholarship , School of Software Engineering, Sun Yat-sen University	2023

SKILLS

Programming : C/C++, CUDA, Python
Frameworks & Tools : llama.cpp, GGML, vLLM, SGLang, PyTorch, NVIDIA Nsight Systems, NVTX, LaTeX

LANGUAGE PROFICIENCY

IELTS Academic : Overall 7.5